

Training, Fine-Tuning, and Customization

ML Model and Foundation Model Development

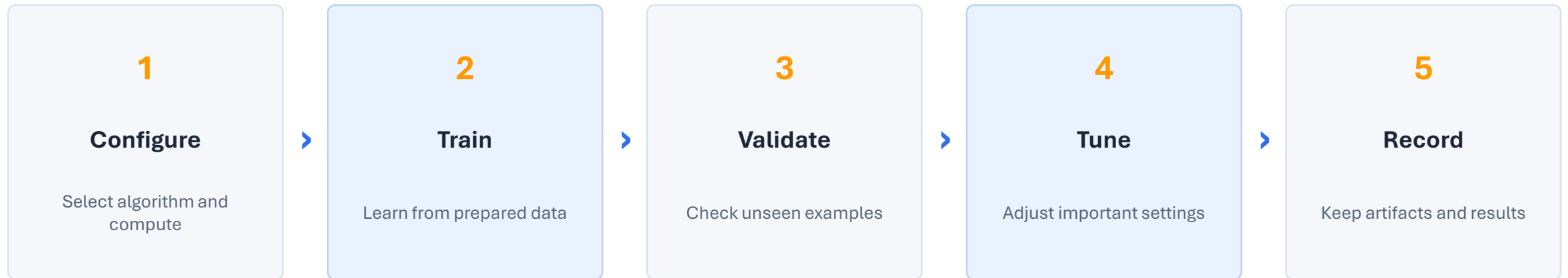
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Model choices • Training • Evaluation



Training turns a model choice into a working system

A controlled process connects prepared data to a reproducible model.



SageMaker AI supports several training approaches

The right option depends on control, speed, and framework needs.

Built-in algorithms

Use managed implementations for common machine learning tasks.

Script mode

Bring training code with supported frameworks such as PyTorch or TensorFlow.

Custom containers

Package special libraries or environments when standard options do not fit.



Hyperparameters control how learning happens

They shape training behavior but are not learned directly from the data.

Learning settings

Learning rate, epochs, steps, and batch size affect convergence.

Model settings

Depth, regularization, and architecture affect capacity.

Automatic tuning

SageMaker automatic model tuning compares trials using a chosen metric.



Underfitting and overfitting need different responses

Validation behavior shows whether the model learned too little or memorized too much.

Underfitting

- Weak results on training and validation data
- Model may be too simple
- Features may be unhelpful
- Train longer or increase capacity

Overfitting

- Strong training results but weak validation
- Model may be too complex
- Data may be limited or noisy
- Regularize, simplify, or add data



Training efficiency reduces time and cost

Optimization techniques should preserve the required model quality.

Early stopping

Stop when validation performance no longer improves.

Distributed training

Use multiple workers when the dataset or model requires more compute.

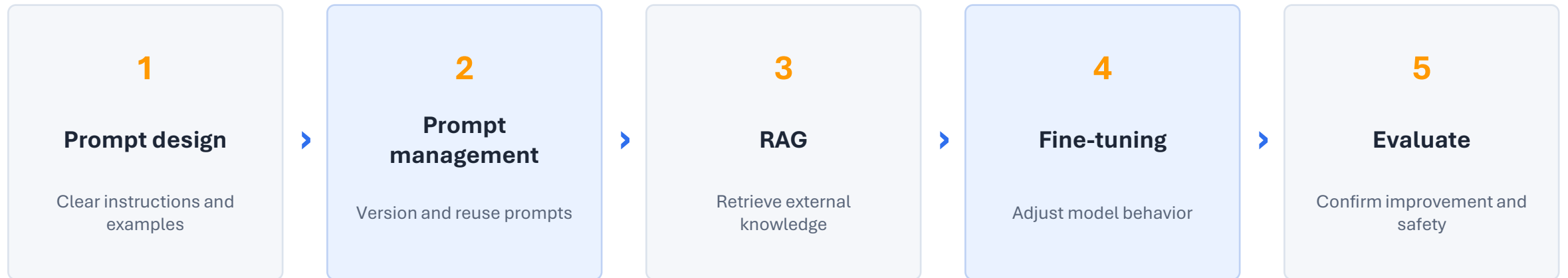
Checkpoints and warm starts

Save progress and reuse earlier training when appropriate.



Foundation models offer several customization levels

Each method changes cost, effort, and control.



Training should stay reproducible

A result has value only when the team can explain and repeat it.

- 1 Version the data, code, prompts, and configuration.
- 2 Record hyperparameters, metrics, compute, and random seeds.
- 3 Separate training, validation, and test data correctly.
- 4 Watch for overfitting, leakage, and catastrophic forgetting.
- 5 Compare quality gains with training time and cost.
- 6 Keep the model artifact and evaluation evidence together.

